Chapter - 1 Introduction To Economics

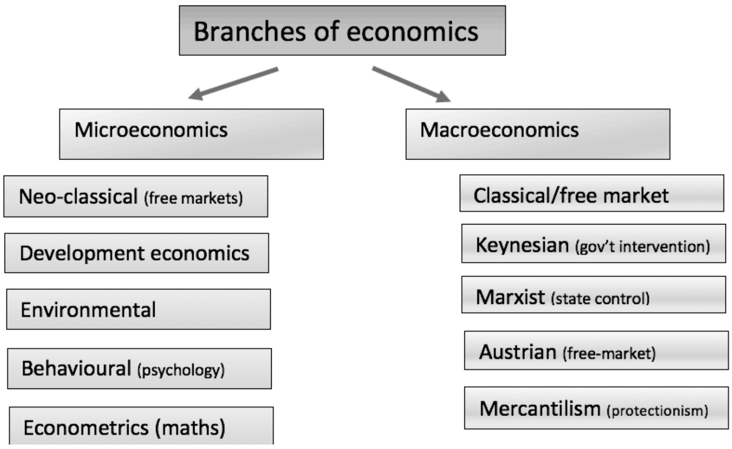
1.1 Economics : Macro And Micro

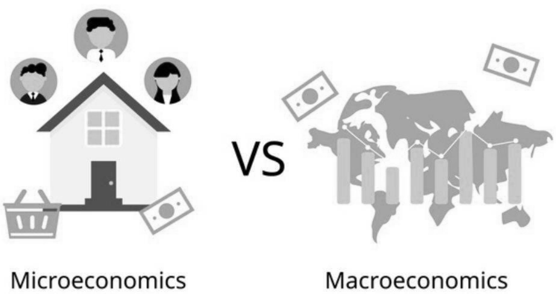
Economics is the study of how to allocate limited resources to unlimited wants .

Adam Smith (1776) - Economics is study of wealth .

Lionel Robbins (1932) - Economics is a science which studies human behaviour as a relationship between ends and scare means which have alternative uses .

Alfred Marshall (1980) - Economics is on one side , study of wealth and in other side study of man .





[2082 Chaitra]

Q. Distinguish between microeconomics and macroeconomics, also explain why an engineer focuses primarily on the former during project evaluation .

Micro Economics	Macro Economics
Studies individual economic units.	Studies the economy as a whole.
Deals with individual prices, demand, supply, production, and costs	Deals with GDP, inflation, unemployment, and economic growth.

An engineer mainly focuses on microeconomics because project evaluation involves a specific project or firm .

1.2 Engineering Economics : Application , Principles And Need

Engineering Economics is the application of economic principles and mathematical techniques to evaluate engineering projects.

Engineering Economics = Engineering + Economics + Decision Making

Engineering Economy is a branch of engineering that deals with the systematic evaluation of costs and benefits of proposed engineering projects .

Its scope covers five key areas:

1. Project Appraisal and Capital Investment Decisions
2. Cost Estimation and Budgetary Control
3. Replacement Analysis of Assets
4. Make-or-Buy Decisions
5. Equipment Selection and Utilization

The Seven Principles Of Engineering Economics Are : **[MOST ASKED]**

1. Develop the alternatives.
2. Focus on the differences.
3. Use a consistent viewpoint.
4. Use a common unit of measure.
5. Consider all relevant criteria.
6. Make risk and uncertainty explicit.
7. Revisit your decisions.

Importance of economics for engineers:

1. Helps in selecting the most economical design or project.
2. Assists in cost estimation and budgeting.
3. Enables efficient allocation of resources (materials, labor, and capital).
4. Supports investment and replacement decisions for machines and equipment.
5. Improves profitability and productivity while reducing unnecessary costs.

Importance of Engineering Economics:

1. Helps compare different engineering alternatives.
2. Ensures efficient use of scarce resources such as money, materials, labor, and time.
3. Assists in cost-benefit and life-cycle cost analysis.
4. Evaluates the economic feasibility of projects.
5. Minimizes production, operation, and maintenance costs.

1.3 Role Of Engineering In Decision Making [MOST ASKED]

Engineers need knowledge of economics because resources such as money, materials, time, and labor are scarce.

Economic principles help them choose the most efficient and cost-effective alternative among different engineering options.

- Compare different project alternatives.
- Minimize costs and maximize benefits.
- Allocate scarce resources efficiently.
- Evaluate investment and project feasibility.
- Make decisions based on economic analysis rather than technical factors alone.

Roles of an engineer in economic decision-making:

1. Identifies and compares alternatives to solve engineering problems.
2. Performs cost-benefit analysis to select the most economical option.
3. Uses scarce resources efficiently such as money, materials, labor, and time.
4. Evaluates project feasibility before implementation.
5. Minimizes production, operation, and maintenance costs.

1.4 Terms Associated With Economics

Socialistic Economy **[2066 Magh]**

A socialist economy is an economic system in which the government owns and controls the major means of production and distribution for the welfare of society.

Opportunity Cost **[2074 Bhadra] [2069 Bhadra]**

Opportunity cost is the value of the best alternative foregone when a choice is made.

Cash flow diagram **[2076 Baisakh] [2066 Magh]**

Cash flow diagram is a graphical representation of the cash inflows and cash outflows that occur over time in an engineering or financial project. It helps visualize when money is received and when it is paid.

Debt Ratio**[2082 Bhadra]**

Debt Ratio is a financial ratio that measures the proportion of a company's total assets financed through debt. It indicates the level of financial leverage and the company's dependence on borrowed funds.

Price-to-Earnings Ratio (P/E Ratio) **[2082 Bhadra]**

Price-to-Earnings Ratio is a market valuation ratio that measures the relationship between a company's market price per share and its earnings per share. It indicates how much investors are willing to pay for each unit of the company's earnings.

Profit Margin **[2082 Bhadra]**

Profit Margin is a profitability ratio that measures the percentage of revenue remaining as net profit after deducting all expenses. It indicates how efficiently a company converts its sales into profit.

Total Asset Turnover **[2082 Bhadra]**

Total Asset Turnover is an efficiency ratio that measures how effectively a company uses its total assets to generate sales or revenue. A higher ratio generally indicates more efficient use of assets.

If You Are Find This Note Valuable, You Can Support Me ‼

BEI UPDATED SYLLABUS NOTES

eSewa

Saurab Gyawali

9846981640